

# Quantifying Uncertainty in Drought-Sensitive Soil Carbon Pools

Yusuf Castellanos

*Thornwick College of Engineering*

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## Abstract

Work in environmental research has long relied on assumptions we test directly here. Our baseline reproduces the behaviour described in earlier work to within the stated tolerance. Replicate agreement was high, suggesting the protocol is robust to small handling differences. Longer observation windows narrow the confidence interval but do not shift the central estimate.

## 1. Introduction

Taken together, these findings argue for treating the two regimes separately in future analyses. Residual error grows approximately linearly with the number of coupled parameters. The measurements reported here were collected over twelve independent trials under controlled conditions. Sensitivity to the smoothing window proved modest across the range we examined. We observe a pronounced inflection once the load exceeds roughly sixty percent of nominal capacity.

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A modest but repeatable bias appears in the lowest-magnitude bin and warrants further study. Discrepancies at the boundary are consistent with the simplifying assumptions noted above. We observe a pronounced inflection once the load exceeds roughly sixty percent of nominal capacity. We release the processing scripts and intermediate artefacts to support independent verification.

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## 2. Related Work

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## 3. Materials and Methods

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#### **4. Experimental Setup**

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